

MSBA 511: Data Mining for Business Analytics

Conceptual Homework 4: k-Nearest Neighbors

Objective

Be able to:

- Apply the steps of the k-Nearest Neighbors classification algorithm to make a classification based on sample patient data.

Overview

A medical clinic is using the k-Nearest Neighbors (KNN) algorithm to predict whether a patient has a high risk of heart disease (1 = High Risk, 0 = Low Risk) based on their age, BMI, and total cholesterol level. The dataset shows the records of seven patients as well as a new patient that needs to be classified below:

Patient	Age	BMI	Total Cholesterol	Risk Level
A	76	28.87	245	1
B	71	21.63	195	0
C	43	23.71	180	0
D	72	20.37	235	1
E	56	34.33	185	1
F	58	24.37	165	0
G	74	24.68	220	1
New	67	25.4	215	?

Tasks

1. Standardize the Age, BMI, and Total Cholesterol variables. For the purpose of this assignment, we will use a [Min/Max Scaling](#) method since the mean and standard deviation are unknown. Use the following min/max values:

Variable	Min	Max
Age	18	100
BMI	15	45
Total Cholesterol	50	350

2. Calculate the Euclidean distance between the new patient and all 7 patients in the sample data.
3. Identify the **three** nearest neighbors to the new patient based on the distances you calculated.
4. Use majority voting among the **three** nearest neighbors to classify the risk level of the new patient.

Submission Requirements

Submit a PDF document to Gradescope that includes the following:

1. Complete Tables 1 and 2 in the following template:
[HW4_K-Nearest_Neighbors_Template \(7.5 Points\)](#)
2. If **four** nearest neighbors were used to classify the risk level of the new patient, would this change your classification? **(2.5 Points)**
3. Documentation for each step, and all relevant calculations. You may include screenshots or graphics in addition to a textual summary on page 2 of the above template or in some other written format. **(5 Points)**
 - a. **No credit will be given for Part 3 if your submission shows evidence that Python was used for more than validation of your work.**